Software Test Plan (STP) - Online Bookstore System

Project: Online Bookstore System

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1. Introduction

Purpose: This document defines the test plan for the Online Bookstore System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers Online Bookstore features such as customer registration and login, catalogue browsing and search, cart and checkout, online payment, order history and status, and administrator functions (book, category, inventory and order management, reporting). Internal operation of the external payment and delivery services is excluded.

References: Online Bookstore System SRS v1.0 (09-09-2026), Online Bookstore System SAD v1.0 (14-09-2026), Requirements Traceability Matrix (SRS Section 8, SAD Section 5.4).

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design), RTM (Requirements Traceability Matrix), FR (Functional Requirement), NFR (Non-functional Requirement), SR (Security Requirement), TC (Test Case), UAT (User Acceptance Testing).

2. Test Items

- Authentication & Authorization module
- Catalogue module
- Cart & Checkout module
- Order & Payment module
- Inventory module
- Reporting module
- Admin interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

- OBS-F-001 – OBS-F-003: Account creation, login, logout (TC-Auth-01–03)
- OBS-F-004 – OBS-F-008: Catalogue display, search, filter, sort, book detail (TC-Cat-01–04, TC-Book-01)
- OBS-F-009 – OBS-F-011: Cart add/update/remove with stock-aware quantity limit (TC-Cart-01–03)
- OBS-F-017: Stock quantity is not reduced when payment fails; stock is reduced only after successful order confirmation (TC-Cart-04)
- OBS-F-012 – OBS-F-013: Delivery details capture and order total calculation (TC-Order-01–02)
- OBS-F-014 – OBS-F-015: Payment request and verified-result handling (TC-Pay-01–02)
- OBS-F-016: Order confirmation with unique order number (TC-Order-03)
- OBS-F-017: Stock deduction on confirmed order (TC-Inv-01)
- OBS-F-018 – OBS-F-019: Order history, status view, cancellation (TC-Order-04–05)
- OBS-F-020 – OBS-F-022: Book and category management (TC-Admin-01–03)
- OBS-F-023 – OBS-F-024: Stock/reorder-level update and low-stock identification (TC-Inv-02–03)
- OBS-F-025: Administrator order-status update (TC-Admin-04)
- OBS-F-026: Sales and inventory reports (TC-Rep-01)

- OBS-NF-001: Response time ≤ 3s, 90th percentile (TC-Perf-01: measures response times for representative customer operations under normal load; verifies at least 90% of measured responses complete within 3 seconds)
- OBS-NF-002: Monthly availability ≥ 99.5% (TC-Reliability-01: availability monitoring and calculation; full monthly compliance requires a 30-day measurement period beyond this execution window)
- OBS-NF-003: First-time usability, search and add to cart (TC-UX-01)
- OBS-NF-004: Scalability – 10,000 book records / 20 concurrent sessions (TC-Scale-01)
- OBS-NF-005: Cross-device usability, desktop/tablet/mobile (TC-UX-02)
- OBS-NF-006: Administrator audit trail (TC-Audit-01)
- OBS-SR-001 – OBS-SR-005: Authentication, password hashing, RBAC, secure communication, input validation (TC-Sec-01–05)
- OBS-SR-006: Payment-result verification before marking an order paid (TC-Sec-06)

4. Features Not to be Tested

- Internal processing logic of the external Payment Gateway (provider responsibility)
- Internal operation of the external Delivery Service
- Notification service for order confirmation/status (optional, out of scope)
- Underlying database engine, ORM framework, and third-party library internals
- Physical packing and delivery logistics

5. Test Approach / Strategy

Levels:

- Unit tests (module-level)
- Integration tests (service-to-service, and Order & Payment Service ↔ external Payment Gateway sandbox)
- System tests (end-to-end bookstore functionality)
- Acceptance tests (UAT)

Types:

- Functional testing (core features)
- Regression testing
- Performance testing (response time, load)
- Usability testing (UI clarity, accessibility, cross-device)
- Security testing
- Scalability testing (10,000 records / 20 concurrent sessions)

Entry Criteria: Stable build delivered, test data available (seeded catalogue of 10,000 books), payment gateway sandbox reachable, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Authentication/session testing: verify protected functions cannot be accessed without valid authentication
- Validate password storage (salted hash, never plaintext)
- TLS verification for account/order/payment traffic
- RBAC boundary testing between customer and administrator
- Security assessment: penetration testing and fuzzing against authentication and input-validation boundaries (account, search, address, admin forms)
- Payment-result signature verification (tampered results must not mark orders paid)

6. Test Environment

Hardware: Standard client devices – desktop, laptop, tablet, and mobile.

Software: Bookstore application v1.0 (Customer Web UI, Administrator UI), backend REST services, relational database, external Payment Gateway sandbox.

Tools: Selenium (UI automation), Postman (API), JMeter (performance), OWASP ZAP (security), Jira (defect tracking).

Test Data: Dummy customer accounts, seeded catalogue of 10,000 book records, mock payment responses (success/failure/timeout/tampered).

7. Test Schedule

Milestones:

- Test case design: 17-Sep-2026
- Environment setup: 18-Sep-2026
- Test execution start: 19-Sep-2026
- Test execution end: 26-Sep-2026
- UAT: 27-Sep-2026 to 29-Sep-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead		Prepare plan, coordinate execution
Test Engineer		Design & execute test cases, log defects
Test Engineer		Design & execute test cases, log defects
Developer		Support defect fixes and triage
Product Owner		Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
External Payment Gateway sandbox unavailable or unstable	Use mocked payment responses (success/failure/timeout) so testing is not blocked
Stock incorrectly reduced after failed payment	Add explicit test case (TC-Cart-04) verifying that failed payment does not reduce available stock and that stock remains consistent with confirmed orders
Load test cannot reach 10,000 records / 20 concurrent sessions	Provision a seeded test database and scripted concurrent-session load ahead of execution
Limited team size and timeline	Prioritize high-priority requirements first

11. Assumptions & Dependencies

- SRS v1.0 and SAD v1.0 will remain stable during the test cycle
- Test data (accounts, catalogue, inventory) will be provided before execution
- Payment gateway sandbox/mock will be available and stable

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for >4 hours
- Build is too unstable (blocks >30% test cases)
- Payment gateway sandbox unreachable for an extended period with no mock fallback

Resume testing if:

- Blocking defects are resolved
- Environment stabilized

13. Test Case Management & Traceability

The mappings below provide traceability between selected SRS requirements and their planned test cases.

Traceability mappings:

- OBS-F-002 (Customer login) → TC-Auth-02
- OBS-F-011 (Stock-aware cart limit) → TC-Cart-03
- OBS-F-017 (Failed payment does not reduce stock) → TC-Cart-04
- OBS-F-015 (Payment result verification) → TC-Pay-02
- OBS-NF-001 (Response time) → TC-Perf-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report